PDF Report

Team Name: QuarkStack

Project Name: QuarkSketch

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Module Name: Project Management

Assignment Name: Project Inception

Lecturer: Joseph Corr

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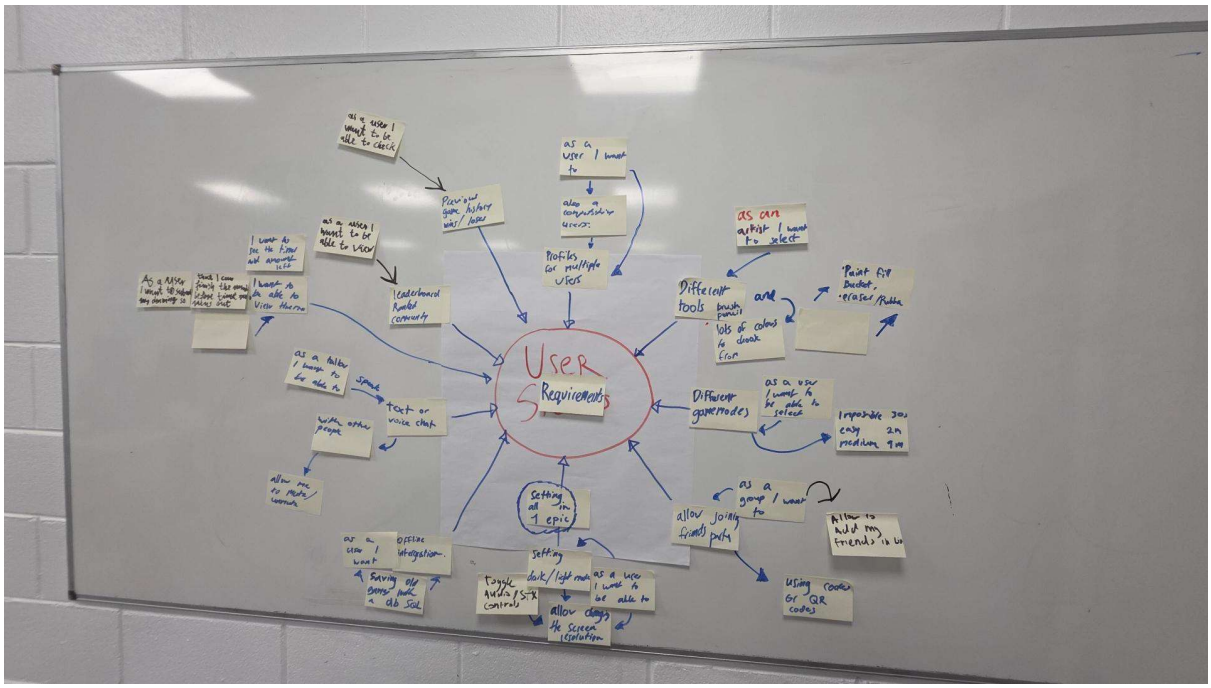
Brainstorm/sketch your project requirements for your complete application



Writeup/Document/ your requirements in User story format

Kamil

1. As a user, I want to see the prompt clearly so that I know what to draw.
2. As a user, I want a countdown timer so that I know how much time is left.
3. As a user, I want a warning in the last 5 seconds so that I can finish quickly.
4. As a user, I want the round to end automatically so that the game is fair.
5. As a user, I want to see my score after each round so that I know my result.
6. As a user, I want to start a new round easily so that I can keep playing.
7. As a user, I want the canvas to be responsive so that drawing feels smooth.
8. As a user, I want a submit button so that I can finish early.
9. As a user, I want to see which round I am on so that I know my progress.
10. As a user, I want clear buttons so that the app is easy to use.

Niall

1. As an artist I want to be able to select different brushes so I can have different sizes
2. As a user I want to be able to select different gamemodes to have a variety to play
3. As a talker I want to be able to speak with voice chat to communicate
4. As a typer I want to be able to type with chat to communicate
5. As a player, I want a timer for each round so that the game keeps moving
6. As a player, I want an eraser tool so that I can correct mistakes.
7. As a player, I want to clear the canvas so that I can restart my drawing if needed.
8. As a player, I want an undo button so that I can quickly fix errors.
9. As a player, I want to see who is currently in the lobby so that I know when everyone has joined.
10. As a user, I want simple instructions so that new players can understand the rules quickly.

Daniel

1. As a user, I want to pause before the round starts so that I can prepare.
2. As a user, I want a countdown before the game begins so that I am ready.
3. As a user, I want the timer to be visible at all times so that I can manage time.

4. As a user, I want a results screen so that I can review my performance.
5. As a user, I want the app to load quickly so that I can start fast.
6. As a user, I want to retry a round so that I can improve my drawing.
7. As a user, I want the prompt to change every round so that it stays interesting.
8. As a user, I want drawing to stop when time ends so that rules are enforced.
9. As a user, I want visual feedback when time ends so that I know the round is over.
10. As a user, I want the interface to be simple so that it is easy to learn.

Marko

1. As a user, I want to be able to enable and disable SFX/Music Elements
2. As a user, I want to able to enable/disable Dark Mode
3. As a user, I want to view history of drawings that I've been participating and drawing
4. As a user, I want to create user profile and be able to add friends
5. As a user, I want to join games using QR code/Random generated PIN code
6. As a user, I want to be able to create games and decide my own choice of topic for the group
7. As an artist, I want to save my drawings when a game is done.
8. As a user, I want to have accessibility options (i.e change text font)
9. As a user, I want the ability to stop or restart a match lobby.
10. As an artist, I want the ability to draw using touch screen (for devices/PCs with touchscreen capabilites)

Group your stories into Epics

EPIC 1 – Core Gameplay & Game Flow

Description:

This epic covers the main game loop including rounds, prompts, timer, and scoring.

User Stories:

- Kamil 1 – See prompt clearly
- Kamil 2 – Countdown timer
- Kamil 3 – Warning in last 5 seconds

- Kamil 4 – Round ends automatically
- Kamil 5 – See score after each round
- Kamil 6 – Start new round easily
- Kamil 8 – Submit button
- Kamil 9 – See which round I am on
- Niall 5 – Timer for each round
- Daniel 1 – Pause before round
- Daniel 2 – Countdown before game begins
- Daniel 3 – Timer visible at all times
- Daniel 4 – Results screen
- Daniel 6 – Retry round
- Daniel 7 – Prompt changes every round
- Daniel 8 – Drawing stops when time ends
- Daniel 9 – Visual feedback when time ends

EPIC 2 – Drawing System & Canvas Tools

Description:

This epic includes all drawing functionality and canvas interaction.

User Stories:

- Kamil 7 – Responsive canvas
- Niall 1 – Select different brushes
- Niall 6 – Eraser tool
- Niall 7 – Clear canvas
- Niall 8 – Undo button
- Marko 7 – Save drawings
- Marko 10 – Draw using touchscreen

EPIC 3 – Multiplayer, Lobby & Communication

Description:

This epic covers multiplayer features, lobby management, and communication.

User Stories:

- Niall 3 – Voice chat
- Niall 4 – Text chat
- Niall 9 – See who is in lobby
- Marko 5 – Join games via QR / PIN
- Marko 6 – Create games and choose topic
- Marko 9 – Stop / restart lobby

EPIC 4 – User Interface & Usability

Description:

This epic focuses on usability, navigation, accessibility, and general interface.

User Stories:

- Kamil 10 – Clear buttons
- Niall 10 – Simple instructions
- Marko 2 – Dark mode
- Marko 8 – Accessibility options
- Daniel 5 – App loads quickly
- Daniel 10 – Simple interface

EPIC 5 – User Accounts, Progress & Social Features

Description:

This epic includes profile management, saving progress, and social features.

User Stories:

- Marko 3 – View history of drawings
- Marko 4 – Create user profile and add friends

EPIC 6 – Audio & Game Customisation

Description:

This epic covers optional features that improve user experience.

User Stories:

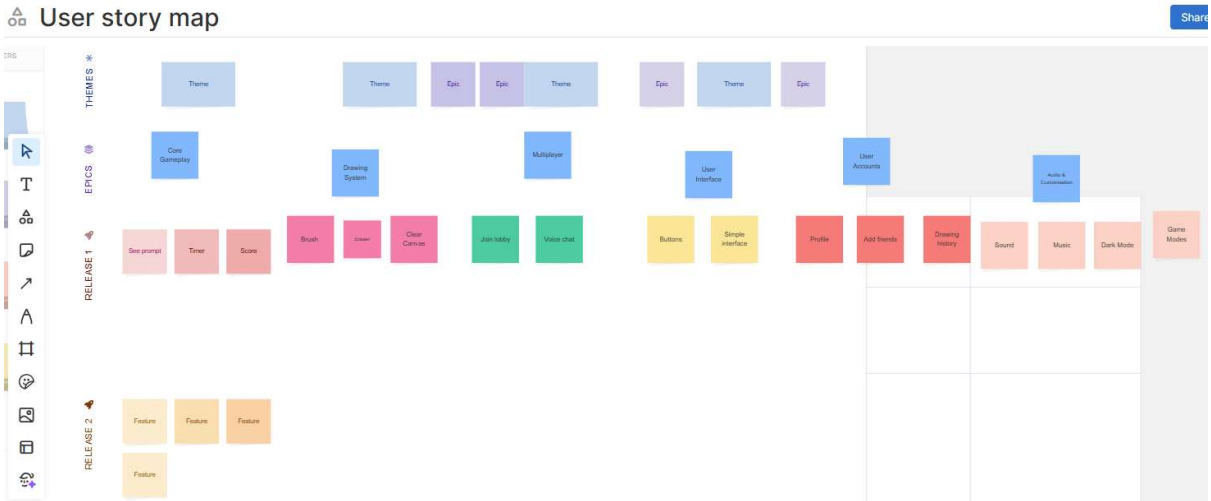
- Marko 1 – Enable / disable SFX and music
- Niall 2 – Select different game modes

Add your user stories and epics to your Jira project

https://atu-team-sa3.atlassian.net/jira/software/projects/SCRUM/boards/1

Create a user story map to group related stories.

https://quarkstack.aha.io/pages/PROD-N-1?tour_type=whiteboards



Brainstorm and document your team's definition of Ready (DOR)

A backlog item is "ready" when:

- It is clearly written and understood.
- Acceptance criteria are defined.
- It is small enough to finish in one sprint.
- It has been estimated.
- Dependencies are identified.
- It is prioritized by the Product Owner.

Brainstorm and document your team's definition of Done (DOD)

A backlog item is "done" when:

- The work is completed.
- Code is reviewed.
- It meets all acceptance criteria.
- It is tested and works as expected.
- No critical bugs are present.
- It is integrated into the main branch/build.
- Documentation is updated.
- The Product Owner has reviewed and accepted it.

Technical Architecture

Frontend

- Built with Angular (TypeScript)
- Provides the drawing canvas and guessing interface
- Sends and receives data from the backend using HTTP requests

Backend

- Built with Node.js and TypeScript
- Handles game logic (rounds, prompts, scoring)
- Manages players and game sessions
- Communicates with the database

Database

- MySQL database
- Stores:
    - Users
    - Game rooms
    - Prompts
    - Scores
    - Game history

Define/ sketch your development environment and tool requirement

Programming Language

- TypeScript

Framework

- Angular (Frontend)
- Node.js (Backend)

Database

- MySQL

Development Tools

- VS Code
- Git & GitHub
- MySQL Workbench

Define/ sketch your testing strategy, testing requirements and tools to be used

Marko will test the application in the following ways:

- **Unit Testing**
  Test individual components and functions to make sure they work correctly.
- **Integration Testing**
  Test that the frontend, backend, and database work together properly.
- **Manual Testing**
  Play the game with users to check:
  - Drawing works correctly
  - Scores update properly
  - Game rounds function as expected
  - No major errors occur

All important bugs will be fixed before deployment.

Define/ sketch your deployment environment and tools

The application will be deployed online so users can access it through a web link.

- The Angular frontend will be built for production and hosted online.

- The Node.js backend will be hosted on a server.
- The MySQL database will store the game data.

After deployment, we will test the live version to make sure everything works correctly.